ELC 458_ S2021

1) clock period : $T - T_s \geq t_{cq} + t_{pd} + t_{su}$
hold time : $t_{cq} + t_{pd} \geq t_{hold} - T_s$

2) As the benefit of negative skew is considerable for the hold time to meet it's constraints, while the benefit of positive skew for setup time has a minor effect on meeting setup timing constraints.

3) As for the hold time, both registers are concerned with the same ~~edge~~ clock edge.

4) $T - 2T_j \geq t_{cq} + t_{pd} + t_{su}$

5) High area and power consumption

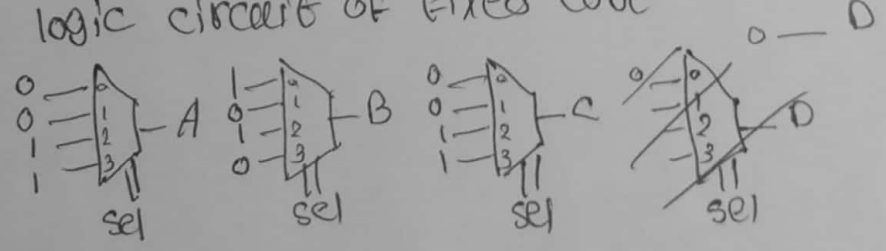
6) Design dependent (Highly stiched to the design)

7) Fixed code → we will assume zero values in non-stated cases

```
process (sel)
begin
  if (sel="00") then
    A<='0'; B<='1'; C<='0'; D<='0';
  Elsif (sel="01") then
    A<='0'; B<='0'; C<='0'; D<='0';
  Elsif (sel="10") then
    A<='1'; B<='1'; C<='1'; D<='0';
  Else
    A<='1'; B<='0'; C<='1'; D<='0';
  End if;
End process;
```

the original code is bad as it will force the synthesizer to infer latches for both B and D to store the latest value in the cases in which they have no defined value

logic circuit of fixed code



8) $N = 40$, $K = 8$, $P = 5$

RCA: $t_{pd})_{RCA} = (N-1)\, t_c + t_{sum}$

$= 39 \times 18 \times 0.69 R_0 C_0 + 30 \times 0.69 R_0 C_0$

$\boxed{= 732 \times 0.69 R_0 C_0}$

carry-by pass:

$t_{pd})_{bypass} = K t_c + (P-1)\, t_{mux} + (K-1)\, t_c + t_{sum}$

$= (2K-1)\, t_c + (P-1)\, t_{mux} + t_{sum}$

$= 15\, t_c + 4\, t_{mux} + t_{sum}$

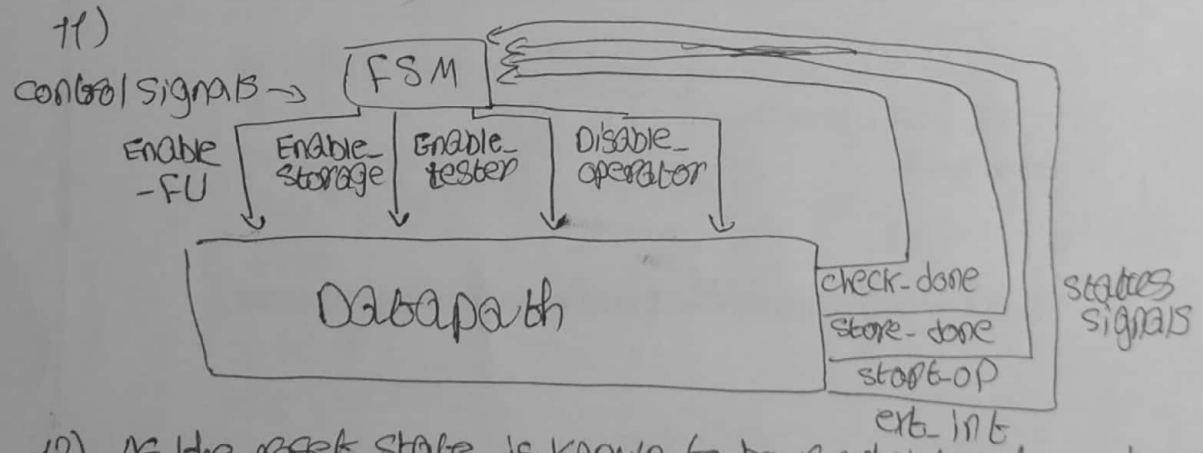
$\boxed{= 316 \times 0.69 R_0 C_0}$

perc. improvement $= \frac{732 - 316}{732} \times 100 = 56.83\ \%$

9) Read and write pointers can be gray encoded which allows each transition to be only 1-bit transition which can pass safely through a dff synchronizer with delays which can be handled through the full and empty flags to ensure proper operation.

10) Keeping wires thick increases interwire capacitance (cross-talk) as it will result in lateral surface area increasing.

11)



12) As the reset state is known to be reachable through an asynchronous reset and this transition will have a higher priority over any other transition

13)
```
process (start_op, ext_int, check_done, store_done,
        current_state)
Begin
  case (current_state) is
   when reset => next_state <= wait;
   when wait =>
    if (start_op = '1') then
      next_state <= operate;
    else
      next_state <= wait;
   when interrupt => next_state <= wait;
   when operate =>
    if (ext_int = '1') then
      next_state <= interrupt;
    else
      next_state <= store;
   when store =>
    if (store_done = '1') then
      next_state <= check;
    else
      next_state <= store;
   when check =>
    if (check_done = '1') then
      next_state <= wait;
    else
      next_state <= check;
   when others =>
      next_state <= reset;
   End case;
  End process;
```

14) $R_{\substack{global\\wire}} = \frac{L}{\sigma A} = \frac{L}{\sigma w t_w} \rightarrow \boxed{R_{\substack{new\\global\\wire}} = \frac{L K_w}{\sigma \frac{w}{K} \frac{t_w}{K}} = K^2 K_w R_{\substack{global\\wire}}}$

15) $\boxed{P_{\substack{global\\wire}} = \frac{V^2}{R} \longrightarrow P_{\substack{new\\global\\wire}} = \frac{1}{U^2} \times \frac{1}{K^2 K_w} P_{\substack{global\\wire}}}$

16) the receiver can lower ack signal when it receives a sychronized lowed req signal, which indicates that the transmitter recognized the high ack signal which implies successful transmission. while lowering it without this sequence cannot ensure proper operation

17)

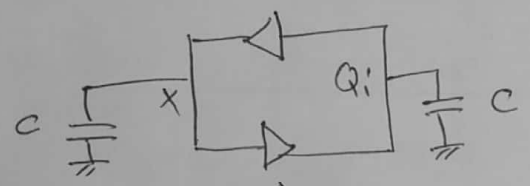
$$t_{pd} = K t_c + \left(\frac{N}{K} - 1\right) t_{mux} + (K-1) t_c + t_{sum}$$

$$\frac{\partial t_{pd}}{\partial K} = 2 t_c - \frac{N}{K^2} t_{mux} = 0$$

$$\text{so } K^2 = \frac{N}{2} \frac{t_{mux}}{t_c} \rightarrow \boxed{K = \sqrt{\frac{N}{2} \frac{t_{mux}}{t_c}}}$$

18) This will increase number of bits per block for lesser delay, as the cost of multiplexer delay will require less usage of it where longer carry chains can have better delay than using multiple muxs.

19)



$$g_m = +(g_{mn} + g_{mp})$$

$$\therefore -g_m V_{Q_i} = C \frac{dV_x}{dt} \quad \&$$

$$-g_m V_x = C \frac{dV_{Q_i}}{dt}$$

$$\therefore g_m (V_x - V_{Q_i}) = -C \frac{d(V_x - V_{Q_i})}{dt}$$

$$\therefore g_m V_d = -C \frac{dV_d}{dt} \rightsquigarrow \tau = \frac{C}{g_m}$$

$$\boxed{\therefore V_d = V_d(0) e^{t/\tau}}$$

$$\therefore V_{DD} = V_d(0) e^{T_{exit}/\tau}$$

$$\boxed{\therefore T_{exit} = \tau \ln\left(\frac{V_{DD}}{V_d(0)}\right)}$$

20)

Entity: it is a black box to be defined in the work library showing only port details: directions, width, names

Component: It ~~is~~ shows the types of blocks used inside our design which can be instantiated once or more than once.

Instance: it is the copy of a component that can be used multiple times in the same design.

the differences between the entity and component is based on the scope, as a higher level entity can see a lower level entity as a component, we can look at the component as the blocks which can be used in the current entity, this usage can be considered as the instance, such that we can have in our entity one type of 8-bit adder which is ~~instantiated~~ used 3 times which implies that we have 3 adders instances.

CamScanner